

Program final update day ACP			26 th June 2019	
<i>Location: Ruppert building, room 0.42</i>				
From	To	UU supervisor		Posters
9:15	9:30	Welcome (and setup first speakers)		Bas de Vos
9:30	9:45	Henrike Haase	Krista Overvliet	Henrike Haase
9:45	10:00	Alex Trout	Nico Romeijn	Stefan Gaillard
10:00	10:15	Roderic Hillege	Nico Romeijn	Anne Reuten
				Annemieke San Giorgi
				Bas Oppenheim
10:15	10:30	Break		Eefje Kooij
				Emma Castellsaguer Petit
10:30	10:45	Fabienne Heijn	David Terburg	Frits van Iddekinge
10:45	11:00	Floor Elbertse	Stella Donker	Ime Dekker
11:00	11:15	Inge Koehorst	Sita ter Haar	Jan Schulp
				Jasmine Gerritsma
11:15	11:30	Break		Jochem Kolsteeg
				Joran Westerink
11:30	11:45	Mats Bakker	Martijn Mulder	Judith Kujiper
11:45	12:00	Kazuya Takahashi	Jeroen Benjamins	Laura Osinga
12:00	12:15	Joy Zwart	Martijn Mulder	Mila van der Zwaag
				Mitchel Kappen
12:15	13:15	Lunch break		Nathaniël Haanschoten
				Niilo Valtakari
13:15	13:30	Layla Liu	Stella Donker	Rob Koppenaar
13:30	13:45	Margreet Wijmans	Ignace Hooze	Sjors van de Ven
13:45	14:00	Mona Zimmermann	Ben Harvey	Sophia Eijssman
				Willem de Bruijne
14:00	14:15	Break		Willem van der Maden
14:15	14:30	Joel Tan	Joris Elshout	
14:30	14:45	Thomas Camminga	Marloes van Dijk	
14:45	15:00	Piet Jonker	Joris Elshout	
15:00	15:15	Break		
15:15	15:30	Stefan Gaillard	Jeroen Benjamins	
15:30	16:00	Pitches		
16:00	± 17:30	Posters & drinks		

Presentation abstracts

Name: Henrike Haase

Project Title: How people with a visual field defect scan their environment: An Eye-Tracking Study

Abstract: The scanning behavior of people with a Visual Field Defect (VFD) is still relatively unexplored, although a VFD can have great impact on the person concerned with it. This study examined the scanning behavior of people with a VFD in a mobility situation by the means of an Eye-Tracker. Participants were asked to watch a video of everyday mobility situations, showing both walking and biking sequences from first person perspective. They were asked to imagine being that person in the video and look around naturally. In the video, 33 objects were defined as regions of interest (ROIs), as they are important to detect in order to walk or bike safely. The Eye-Tracker recorded when and where the participants looked. It was expected, that people with a visual field defect (VIPs) need more time to fixate on relevant ROIs for the first time, detect less ROIs in total, do not look at them as long as sighted people do and fixate more often on the ROIs. The results confirmed that VIPs detect less ROIs and spend less time looking at them. Contrary to what was expected, the results also showed that sighted people fixate more on the ROIs than VIPs. An additional analysis showed that the time until an ROI is first looked at, is significantly shorter for sighted people. These findings showed that there are indeed differences in scanning behavior and they could be used to help VIPs counterbalance their decreased visual field.

Location project: Royal Dutch Visio, Amsterdam

External supervisor: Jan Koopman

UU supervisor: Krista Overvliet

Name: James A. Trout

Project Title: Exploring Experience Level and UI use in StarCraft 2.

Abstract: As software becomes more and more demanding of our time and cognitive faculties, how this software is designed is becoming more and more important. For my thesis I am examining how the needs of users change depending upon their level of experience with a piece of software; specifically for this study, StarCraft 2. I will be using eye tracking and gaze mapping to determine what elements of the user interface (UI) is used in each of three groups: those with no experience, those with some experience, and those with a lot of experience with StarCraft 2. By examining what areas of interest are looked at and for how long for each group and comparing the results, I hope to determine what information is most important to each group, and how the needs of different users differ. Hopefully by using these results, design companies can create interfaces and settings that can cater to the needs of users at different levels of experience.

Location project: Utrecht University

External supervisor: N/A

UU supervisor: Nico Romeijn

Name: Roderic Hillege

Project Title:

Abstract:

Location project:

External supervisor:

UU supervisor:

Name: Fabienne Heijn

Project Title: EmojiGrid: A Validation Study for Immersive Virtual Reality Videos

Abstract: (verbal) Self-report questionnaires are the most widely used tools to assess one's affective response. However, as language-dependent tools are cognitively demanding for the user, specific emotions and feelings are generally hard to explain and lexicons vary across cultures and languages, graphical self-report tools might prove to be more useful. For that reason, Toet et al. (2018) developed the EmojiGrid: a graphical intuitive self-report tool, which does not rely on written instructions.

In their study, the EmojiGrid was validated in context of food-elicited emotions. Participants rated the valence and arousing quality of food images, using either the EmojiGrid or two independent conventional visual analog scales (VAS). Unlike the VAS tool, results obtained with the EmojiGrid yielded the typical U-shaped relation between the mean valence and arousal.

As the emoji used on the EmojiGrid represent facial expressions related to the general affective dimensions of valence and arousal, the EmojiGrid may also prove to be a useful tool to evaluate other affective stimuli such as music, smells and videos. Therefore, this study focused on validating the EmojiGrid in context of immersive Virtual Reality (VR) videos.

Location project: TNO Soesterberg

External supervisor: Alexander (Lex) Toet

UU supervisor: David Terburg

Name: Floor Elbertse

Project Title: Visualising taste: Exploring factors in crossmodal correspondence between taste and shape

Abstract: Recent research has shown that not only do people associate sweet foods with round shapes and other tastes with more angular shapes, there are other attributes which influence this crossmodal correspondence. In an online experiment, these attributes (roundness, symmetry and number of elements) are measured in a shape-taste matching task, using food stimuli of different tastes and intensities. In two online experiments, these crossmodal correspondences have been further examined.

In both experiments, participants were shown four different food stimuli and asked to choose a matching shape along a sliding scale. In experiment 1, no significant effects were found between shape (roundness, symmetry and number of elements) and the sweetness of foods. Taste intensity did have an effect on the number of elements in shapes. In experiment 2, participants were explicitly instructed to match a shape to the expected taste of the shown food stimuli, an instruction that was not present in experiment 1.

Results from experiment 2 show a highly significant effect of expected taste on chosen shape attributes, as well as an effect of taste intensity of the number of elements in chosen shapes. Not only provides this evidence for taste-shape crossmodal correspondences, it also teaches us an important lesson in participant instruction and experiment design.

Location project: TNO

External supervisor: Lex Toet

UU supervisor: Stella Donker

Name: Inge Koehorst

Project Title: Combining FaceReader and eye tracking: measuring the influence of emotions in usability research

Abstract: A website with a low usability can bring out negative emotions and can lead to users' having a negative attitude towards the website. Our assessment of an event is based on the emotions that are elicited during that event and in particular on the emotion we experience last. In usability research, the participant is often asked to perform a search-task. The participant is motivated to fulfill the task, even if the website elicits feeling of frustration because of its bad usability. Eventually the participant will experience the feeling of success, when the task is successfully accomplished. This study examined if successfully accomplishing a task played a moderating role in the relationship between the usability of a website and the attitude towards the website, but no interaction effect was found. Subsequently, this study investigated if measuring facial expressions gives the researcher information about bad design factors in a website that causes usability

problems. FaceReader en eye tracking data were combined to see if participants showed more negative facial expressions when performing a search-task on a website with low usability than in a website with high usability, but no significant difference was found.

Location project: Noldus Information Technology

External supervisor: Patrick Zimmerman

UU supervisor: Sita ter Haar

Name: Mats Bakker

Project Title: Do single AOW pensioners who live abroad, show selfish behaviour against the Netherlands? A comparison between single AOW pensioners living in the Netherlands and single AOW pensioners living in Spain

Abstract: The Social Insurance Bank (SVB) provides the National Old Age Pensions Act (AOW) for elderly who are living in the Netherlands or have lived in the Netherlands. The cost of AOW fraud from elderly abroad is estimated €36,9 million each year. Fraud considering the AOW is more common for elderly living abroad in comparison to elderly who live in the Netherlands. A reason for this could be the social and spatial distance between people who live and the Netherlands and people who live abroad. This distance could be reduced by introducing social norms to the elderly who live abroad. 355 Single AOW pensioners filled in a questionnaire about their feelings towards the Netherlands. This research showed that elderly who lived abroad felt more social distance towards the Netherlands than elderly who lived in the Netherlands. Furthermore, by introducing an experience story which communicated the social norm of not committing fraud, elderly made less self-serving choices.

Location project: Sociale Verzekeringsbank

External supervisor: Femke ten Velden

UU supervisor: Martijn Mulder

Name: Kazuya Takahasi

Project Title: The eye contact effect on cooperative performance:

Abstract: Making eye contact, more broadly the perception of gaze or often termed as social cognition, has been argued as a human-specific mode of communication. It not only functions for humans to transfer their knowledge but also to show their emotions and trust. Moreover, recent studies have revealed that eye contact plays an important role to send a signal of commitment. We hypothesize that making eye contact helps people work successfully together for shared goals and investigated it by using wearable eye tracking glasses. A comparison of the result in Measurement 1 and Measurement 2 indicates that making eye contact does contribute to the increase in performance.

Location project: Amsterdam Japanese School

External supervisor: N/A

UU supervisor: Jeroen Benjamins

Name: Joy Zwart

Project Title: Handheld vs handsfree texting as predictors of dangerous driving behavior.

Abstract: Background: Research has overwhelmingly found that mobile phone use can degrade the range of driving performance measures. Most of the previous research focussed on handheld texting while driving (increases crash risk by 347%). At the moment, hand held phone use while driving is illegal. Handsfree phone use, however, is not.

Method: Using a driving simulator we let 28 participants drive the same route on a highway; once without a phone, once with a phone in their hand and once with a phone mounted onto the dashboard. During the

drives, the participants had to reply to texts. Their gaze directions were recorded with eye tracking glasses. Results: Lateral sway was significantly higher in the conditions with a phone compared to the baseline condition, there was no significant differences between the handheld and handsfree conditions. The same was found for consistency of speed. The average time of glances on the phone (off the road) are significantly longer in the handsfree condition compared to the handheld condition. The same was found for the amount of times glanced at the phone.

Conclusion: Texting with your phone in a holder is just as dangerous as it is with the phone in your hand, road awareness is even worse when the phone is in a stand.

Location project: SWOV

External supervisor: Willem Vlakveld

UU supervisor: Chris Janssen

Name: Layla Liu

Project Title: The Network Analysis for Odor-Evoked Affect at Individual Level, Including Threshold, Current Mood, Perceived Intensity, and Familiarity

Abstract: Odors have been shown to elicit affective responses, but the underlying mechanisms and the individual differences there remain unclear. We hypothesize four individual factors (i.e. current mood, olfactory detection threshold, perceived intensity, and familiarity) influence odor-evoked affect. Network analysis is applied to effectively investigate the relations between these variables. After reporting their current mood, 52 participants rated olfactory detection threshold, valence, arousal, perceived intensity and familiarity of 40 different odors. We used the graphical LASSO approach to explore partial correlations between variables in the network of unpleasant and pleasant odor-evoked affect respectively, investigated the node centrality in each network to find out the mediator in our theoretical model, and evaluate the robustness of estimated networks to support our findings. The results suggest that (1) Arousal in current mood influenced odor-evoked affect through its impact on perceived intensity. But especially for pleasant odors, current mood could also directly influence odor-evoked affect without any mediator; (2) Valence of current mood influenced threshold but does not have impacts on either perceived intensity or odor-evoked affect; (3) Familiarity has impacts on both perceived intensity and odor-evoked affect; (4) Perceived intensity mediated the relations between odor-evoked affect for unpleasant odors, while odor-evoked affect for pleasant odors was more likely to be directly influenced by subjective factors (i.e. current mood, perceived intensity, familiarity). The limitations and implications are discussed further.

Location project: TNO

External supervisors: Lex Toet, Tanja Krone

UU supervisors: Stella Donker, Marnix Naber

Name: Margreet Wijmans

Project Title: Extreme faces: an eye-tracking study

Abstract: Most research done on face perception makes use of a standardized stimulus set that has been stripped of all possible distractors. When averaging the time people looked at different parts of the face, a triangular viewing pattern can be found, where most time is spent looking at the eyes, then the nose and then the mouth. In this research we try to see if this same pattern can be seen when people look at faces with distractors (e.g. body modifications, injuries, etc.) and what happens to the gaze patterns when you divide the images into categories such as pretty vs ugly and scary vs not scary.

Location project: Utrecht University

External supervisor: N/A

UU supervisor: Ignace Hooge

Name: Mona Zimmermann

Project Title: Exploring the use of digital interventions in the chronic phase of traumatic brain injury: A survey study

Abstract: Acquired brain injury (ABI) is any form of damage to the brain that happened after birth (De Luca, Calabro, & Bramanti, 2018). In traumatic brain injury (TBI), this damage occurs through external injury for example caused by a traffic accident or a fall (De Luca, Calabro, & Bramanti, 2018; Peeters et al., 2015). TBI affects an estimated 69 million people globally per year and is frequently associated with reduced employment and satisfaction with life (SWL)(Ahman et al., 2013; Dewan et al., 2018; Emanuelson et al., 2003; Juengst et al., 2015; McMahon et al., 2014; Olver, Ponsford, & Curran, 1996; Rozenbeek, Maas, & Menon, 2013; Reid-Arndt, Nehl, & Hinkebein, 2007). TBI can persistently affect cognitive and emotional functions in a negative manner (Draper, & Ponsford, 2008; Ruttan et al., 2008; Schretlen et al., 2003). Frequently, persistent problems are especially noticed by patients in the chronic phase, when returning to their daily routines (Dikmen, Machamer, Fann, & Temkin, 2010; Draper, & Ponsford, 2009; Olver, Ponsford, & Curran, 1996; Ponsford, Draper, & Schönberger, 2008; Powell, Machamer, Temkin, & Dikmen, 2001). However, in this stage, TBI survivors might not always get the help they need to tackle these problems (Heinemann, 2002; Ribbers et al., 2007). One possible solution to this, could be the use of digital interventions, specifically applications on smartphones and tablet, to target persistent cognitive and emotional problems (Kettlewell et al., 2018; Linden, Hawley, Blackwood et al., 2016; Evald, 2018; Jak, Seelye, & Jurick, 2012). This study set out to further examine and discuss the use of digital interventions in the recovery process of TBI, especially in the chronic phase. The main question to be answered in this study was: To what extent and how could digital interventions such as smartphone or tablet apps be used for the rehabilitation of cognitive and emotional problems after TBI? An online questionnaire was distributed to TBI patients in the Netherlands and the USA.

Location project: Philips Research Eindhoven

External supervisor: Laura Klaming

UU supervisor: Ben Harvey

Name: Joël Tan

Project Title: The effect of contrast dissimilarity on target detection and identification in peripheral vision

Abstract: Head-up displays (HUDs) are used for military and private use. Although designed to counter distraction, this may be the problem it causes through crowding. Crowding is the deterioration of target-object recognition when visual targets are flanked by other objects. This study examines the effect of contrast dissimilarity on performance in detecting and identifying target-objects in crowded conditions. The effect of target and flanker contrast on the amount of detected and identified targets is measured throughout the experiment and statistically compared. Results show significant differences in each condition with the control condition (uncrowded condition). Besides that, predictability of target contrast does not seem to have an effect on performance. In conclusion, flankers have major influence on detecting and identifying (low contrast) targets in peripheral vision. The opposite applies to predictability. When using HUD systems, users should keep in mind the deleterious effect of high contrast symbology on detecting and identifying faint targets.

Location project: TNO Soesterberg

External supervisor: Frank Kooi

UU supervisor: Joris Elshout

Name: Thomas Camminga

Project Title: Finding inspiration: Is the relationship between bilingualism and divergent thinking moderated by direction of cognition?

Abstract: Studies show that bilinguals outperform monolinguals in divergent thinking tasks. However, what is underemphasized in these studies is the role of environmental stimulation in the creative process. The present

study investigated whether the direction of cognition (i.e., whether attention is directed to the external environment or towards internal information like thoughts and memories) and bilingualism predict creativity scores. Furthermore, it investigated if the relationship between bilingualism and divergent thinking is moderated by the direction of cognition. Twelve bilingual and 32 monolingual Dutch primary school children completed a situated version of the alternative uses task (Guilford, 1967). They were asked to think of alternative uses for a given target object (e.g., a pen), that was surrounded by six additional objects that represented the external environment. Two parameters of divergent thinking were considered: flexibility (i.e., the diversity of the ideas) and originality (i.e., the uniqueness of the ideas). Eye-movements were tracked to assess the amount of time the participants spent averting their gaze, as an indicator of internally directed cognition. Multiple regression analyses revealed that bilingualism and the gaze aversion ratio could not predict flexibility or originality. Moreover, the gaze aversion ratio did not moderate the relationships between bilingualism and the parameters of divergent thinking. The implications of these findings are discussed, as well as future directions that can resolve the questions that were raised by these results.

Location project: Universiteit Utrecht
External supervisor: Marloes van Dijk
UU supervisor: Joke Baas

Name: Piet Jonker
Project Title: Measuring motivation during video interviews using facial psychophysiological markers

Abstract: The traditional employment interview is the most popular recruitment tool, even though it is proven to have bad psychometric properties. Furthermore, cognitive biases result in overconfidence from recruiters in rejecting stigmatized applicants, while undergoing prejudice results in higher cognitive effort in the applicant causing more difficulty in answering questions. Video interviewing is gaining popularity, because of increased fairness, easy access and reduced costs. To increase the fit between job and applicant, an effort was done to use psychophysiological facial information as predictor of job-fit. Students were asked to record their motivation to work at a company during an assessment procedure. Professional recruiters then rated the videos on overall quality, motivation, nervousness and confidence. Then, the model was compared to the raters.

Location project: Neurolytics/Utrecht
External supervisor: Felix Hermsen
UU supervisor: Marnix Naber

Name: Stefan Gaillard

Project Titles:

- 1) Experiencing is active perceiving: the influence of immersive technologies on the experience of autonomous sensory meridian response
- 2) ASMR, synaesthesia and YouTube familiarity: exploratory findings concerning age and autonomous sensory meridian response
- 3) Beyond the audio-visual ASMR paradigm: solely auditory and solely visual triggers of autonomous sensory meridian response

Abstracts:

- 1) There is very little scientific knowledge about ASMR experiences, yet developments with regards to ASMR-inducing videos continue. Recent increases in Virtual Reality technology have enabled the creation of ASMR videos within a VR environment. If one takes into account current literature on the relationship between immersion and intensity, and reported increased immersion in VR environments, it seems plausible that ASMR experiences are more intense in a VR environment as well. This research focused on investigating the experienced intensity in VR with 3D videos and found no significant difference compared to watching videos on a screen. Several fruitful lines of new inquiry were discovered over the course of the experiment.

- 2) Several studies have found a significant correlation between synaesthesia and ASMR. Because synaesthesia is more prevalent in children, one would expect ASMR to be more prevalent in children as well. In addition, children might come into contact with ASMR triggers more often due to more extensive usage of certain platforms such as YouTube. This familiarity might then facilitate the triggering of ASMR. To test these hypotheses, we ran an experiment on a science festival with more than 100 participants with differing age categories. Participants were shown a video meant to trigger ASMR and a control video and subsequently answered multiple questionnaires.
- 3) After multiple new ASMR studies at Utrecht University, it became clear that ASMR was perhaps not only triggered by audio-visual stimuli after all, but also by solely auditory or solely visual stimuli. This hypothesis was investigated by showing participants auditory stimuli, visual stimuli, and audio-visual stimuli, with one condition intended to trigger ASMR and one control condition. ASMR onset was measured using self-report, EDA, and ECG.

Location project: Utrecht University

External supervisor: N/A

UU supervisor: Anouk Keizer, Jeroen Benjamins

Poster abstracts

Name: Henrike Haase

Project Title: Detecting workload using a pressure-sensitive mouse

Abstract: With the rise of complex technical settings in which humans interact with machines on a daily basis, workload on the human operator increases steadily, even to a point, where workload is overwhelming and performance suffers. To prevent this from happening, it is important to assess and predict the users' state in real time to take measures against overload. To this end, machines need to be equipped with technology that is able to adapt to changing user states. Measurements of user states are often intrusive, costly or require specific conditions in order to work properly. Measuring behavior with already available input devices such as a mouse might offer a cheap and comfortable way to identify a user state such as workload. The main focus of this study will be the evaluation of a pressure-sensitive mouse as a possible input device for measuring workload. Additional states will be assessed as well. To induce workload, the Warship Commander Task will be used, which is a quasi-realistic air traffic controller task with varying levels of difficulty. The subjectively perceived level of workload is measured via the NASA-TLX. In every condition, click pressure and duration were assessed. It was shown, that pressure was higher in the more demanding task whereas duration of the mouse click was shorter. This indicated that it is possible to detect a potentially overwhelming level of workload by mouse clicks only, which would offer a much cheaper, easier and less intrusive way than before.

Location project: Fraunhofer FKIE, Wachtberg, Germany

External supervisor: Jessica Schwarz

UU supervisor: Joris Elshout

Name: Stefan Gaillard

See presentation abstract

Name: Anne Reuten

Project Title: Cross-cultural affective food experience

Abstract: Emotional attitudes toward food have recently been discovered as an important factor in consumer behavior. When one is interested in measuring emotions cross-culturally, usage of explicit (visual and verbal) methods is problematic because they are prone to cultural response biases. Implicit (physiological and behavioral) methods could potentially circumvent this issue as they reflect fast, unconscious, and

uncontrollable mechanisms. This study was split into two sessions in which we compared explicit and implicit responses from two cultural groups, Thai and Dutch, towards universal, Thai (Asian), and Dutch (Western) drinks and food pictures. The objective was to investigate whether physiological measures (electrodermal activity and heart rate) and behavioral measures (reaction time when responding to self-report rating scales and sip size) could contribute in comparing affective experiences across cultures objectively, without the cultural biases that affect explicit measures. We hypothesize that explicit responses differ between cultural groups for both universal food due to response biases (H1), and cultural food due to response biases and differences in familiarity level with the food (H2). Contrary to explicit measures, we hypothesize no or small differences between cultural groups exist in implicit responding towards universal food as the food is assumed to be equally familiar to both groups, as biological responding is similar, and as there will be no influence of cultural response biases (H3). We do expect to see a difference between groups in implicit responding towards cultural foods, as we expect 'genuine' differences in emotional experience between the cultural groups due to differences in food familiarity level (H4).

Location project: TNO Soesterberg

External supervisor: Anne-Marie Brouwer

UU supervisor: Sita ter Haar

Name: Annemieke San Giorgi

Project Title: Recommendations on the Design & Discovery of Elastique

Abstract: Elastique is a digital boutique agency, as they describe themselves. Very shortly, they develop digital services such as apps and websites for a wide variety of clients. Recently, the company started improving their creative part of the company, that helps the client optimize or create the concept that they want to have developed. To do this, the company set up a Design and Discovery process, in which 3 sprints are used to get from the outlining of the project to a validated prototype of the concept. I have been giving Elastique recommendations on how to improve this process, with focus on their initial desk research, methods for ideations and validation methods.

Location project: Elastique, Hilversum

External supervisor: Sander Oskamp

UU supervisor: Krista Overvliet

Name: Bas Oppenheim

Project Title: Motivating drivers to interact with a technology-based coaching service

Abstract: While technological expertise is high at SD-Insights, client motivation to interact with their driver coaching service is low. Conceptualised as a general product UX issue, it was determined that product UX should be improved. Key focus points were the means of communication, the automatisisation of generating worthwhile feedback, and the app structure. An algorithm to develop worthwhile feedback was developed, as were many suggestions to improve the app UX, such as structure and discoverability.

Location project: RoboValley, Delft

External supervisor: Jasper Pauwelussen

UU supervisor: Nico Romeijn

Name: Emma Castellsaguer Petit

Project Title: Wayfinding and signage for mobility hubs: a case study at Utrecht Central Station

Abstract: Mobility hubs are places where multiple transportation modes are available and where these different kinds of transportation can be shared. These kinds of commercial, public areas need good wayfinding

strategies to make every traveler's journey as successful and safe as possible. The goal of this thesis is to identify what the information wants and needs are for different travelers. In the first study, 92 travelers were interviewed at Utrecht Central Station to find out what kind of traveler wants what kind of information, what they think of the current signage at the station, and how they would improve this. Results indicated that most travelers had trouble locating themselves, thought signage was not well placed, and had trouble finding different modes of transport, especially bus stations. In a second study, I looked at the questions that were asked at the information booths at the station. The questions correlated with the problems found in the interviews and gave some additional insights. Taken together, this research recommends taking the more vulnerable target groups into consideration, like older adults and people who rarely visit the hub, when designing a future mobility hub.

Location project: Goudappel Coffeng, Deventer

External supervisor: Mariska van Essen

UU supervisor: Chris Jansen

Name: Frits van Iddekinge

Project Title: UX research - Customer Validation Carrousel

Abstract: During my internships at ABN AMRO I have been working on different projects. One of the things I have been doing is coordinating the Customer Validation Carrousel, which was organized each Thursday. At this carrousel, five participants are invite based on varying criteria and five teams get 5x 15 minutes to talk to each participant and validate a topic. Before the carrousel, the teams usually come to our UX research walk-in hour to discuss their ideas. At the walk-in hour, we help them to determine the best way to validate their topic, and we give advice on how to write a test script for the interviews that they are going to do with the participants.

Location project: ABN AMRO

External supervisor: Ike Breed

UU supervisor: Michelle Emons

Name: Ime Dekker

Project Title: Can ValueFlow predict likeability of an advertisement: an investigation of moment-to-moment affective reactions and its relation to marketing metrics

Abstract: This study investigated for commercials the integration of moment-to-moment affective experiences into an overall ad evaluation. The focus was on, whether certain parts: 'begin', 'mid' or 'end' or key elements: 'peak experience', 'final moment' based on the heuristic 'peak-end rule', had more influence in creating a well-liked commercial? But, contradictory with past literature about the peak-end rule, no evidence was found for this. Secondly, an explorative study was conducted to investigate to what extent the moment-to-moment affective experiences transfers to other well-known marketing metrics and whether the ValueFlow does this on a comparable level as overall ad liking. Results show strong relationships with Brand Liking, Call-to-Action and Branding. For the ValueFlow a comparable transfer was found as overall ad liking, indicating further validation of the use of ValueFlow as a valid marketing measurement tool.

Location project: Validators

External supervisor: Erik Prins

UU supervisor: Martijn Mulder

Name: Jan Schulpen

Project Title:

Abstract:

Location project:
External supervisor:
UU supervisor:

Name: Jasmin Gerritsma

Project Title: Retention and Transfer of Cognitive Bias Mitigation Interventions: a Systematic Literature Study

Abstract: Cognitive biases can have adverse decision making. If the decisions made affect others, for instance in policymaking, this is particularly detrimental. One of the current problems policymaking is currently tackling is the energy transition. TNO would like to know if there is convincing evidence that interventions aimed at reducing cognitive biases would be beneficial for policymaking. Numerous such interventions have been developed and evaluated. To be considered useful in the context of policymaking, an intervention would need to retain its effect over a period of time (retention) and outside of the training environment (transfer). A systematic literature study has been conducted to explore what is known about the transfer and retention of bias mitigation interventions. The number of studies addressing more than the effect right after the intervention is limited. Of the studies that do look at retention, some use questionnaires with low reliability. On the bright side, certain biases appeared mitigated 12 weeks after an intervention. There is also some evidence suggesting an intervention influenced people's decisions in daily life. For conclusive evidence, more research is needed.

Location project: TNO Soesterberg
External supervisor: Hans Korteling
UU supervisor: Nico Romeijn

Name: Jochem Kolsteeg

Project Title:

Abstract:

Location project:

External supervisor:

UU supervisor:

Name: Joran Westerink

Project Title: A model says more than a thousand braille dots: is the concept formation of visually impaired people improved by using tactile models?

Abstract: Introduction: In education, pictures are often used alongside text in textbooks. This helps the learning process because it is fun and engaging to look at pictures. In addition, information is encoded in the brain through different channels (dual-coding theory), which improves the formation of concepts. In the absence of sight, pictures can not be used alongside written or spoken texts. Instead, in education of visually impaired people, tools such as tactile drawings and models are often used. Although there has been a lot of research on the subjective experiences of using these tactile tools, very limited research has been on the effect on the formation of concepts when using tactile tools. This thesis thus focused on the question: How does the use of tactile models (tactile pictures and 3d models) contribute to the concept formation of visually impaired people? Methods: seven visually impaired people (Legally - totally blind) were told about three different concepts: The platypus, the hovercraft, and the solar system. During the explanations of the concepts they were presented with either a tactile drawing, a 3d model or nothing. Afterwards, questions were asked about the subject to measure how well the participants understood the concepts. In addition, questions were asked about how the participants experienced the use of the tactile tools and how well they thought they understood the concepts. (Preliminary) results: No differences are expected to be found between the understanding of the concepts when using different tactile tools. In addition, no differences are expected to be found between the subjective measures of concept formation and on how participants experienced the tactile

tools. However, there might be a trend on the perceived increase of knowledge about the concepts. The 3d model condition will have a slightly higher increase on self-judged knowledge about the concepts. When asked which tactile condition works the best for concept formation and should be used in schools, six out of seven participants thought the 3d model would be best. Discussion: No differences were found between the use of tactile drawings, 3d models or nothing. However, participants did report that they liked the 3d models the best and thought that 3d models are the best way to improve learning. The lack of results in the current study might be caused by the way concept formation was measured. Learning is a highly personal process. Standardizing how this is done might lead to big differences in learning between the participants. In addition, the starting point of how much participants already knew about the concepts was very divided. Increasing this knowledge was much easier for certain participants compared to others. This might have reduced the effect of the tactile tools. These two factors, in addition to the low number of participants (n=7) might have heavily influenced the results. Conclusion: Although participants reported that they liked the 3d models very much and thought they worked best, the results did not support these claims. The optimal way to test the effect of the tactile tools would be to use them in an educational setting or in a setting which simulates education as best as possible. The reports and enthusiasm of the participants indicate that it might be worth investigating this on larger scale, so education for visually impaired people might be improved.

Location project: Stichting Accessibility

External supervisor: Eric Velleman /Ruben Brandsma

UU supervisor: Krista Overvliet

Name: Judith Kujiper

Project Title: Should I go should I stay? Research to the underlying processes of estimating time to contact

Abstract: The ability to estimate the time to contact (TTC), which is the time it takes for an object to reach a particular place, is an important quality people need in multiple daily-life situations. Important sensory information used in making judgements based on TTC are speed and distance. But, do these variables have the same effect, or do we have a bias? To investigate this, current research developed a two-alternative forced choice task in which one had to decide which of two objects reached a particular place first. The difference in TTC of these two objects was manipulated based on both distance and speed separately. To be able to analyse the data for both reaction time and accuracy, the drift diffusion model was used. Consequently, it was measured whether the sensory information was equally used in the decision processes for differences in TTC due to distance and due to speed.

Location project: Utrecht University

External supervisor: N/A

UU supervisor: Martijn Mulder

Name: Laura Osinga

Project Title: The impact of Electronic Medical Records on the quality of communication between doctors and patients in the consultation room

Abstract: Abstract – Background: Electronic Medical Records (EMR) are increasingly used in health care organizations and the implementation has changed the dynamics of doctor-patient communication. This transition requires research, since patient-doctor communication might be the most significant component of the healthcare visit, with ramifications for patient satisfaction, adherence to treatment, conflict resolution, laboratory costs and clinical outcomes (Duke et al., 2013, p. 358).

Objective: This article offers insights into the relationship between the amount of eye-contact between physician and patient and the extent to which the physician actively reacts to the cues and concerns of the patient during consultations.

Methods: 24 videotaped consultations of 6 different physicians were used to observe the cues and concerns of patients and the corresponding responses of the physicians. In addition, the eye-movements of physicians were measured with an eye tracker to scale the amount of eye-contact.

Location project: AMC Amsterdam
External supervisor: Chiara Jongerius
UU supervisor: Nico Romeijn

Name: Mila van der Zwaag
Project Title: The success of branded entertainment in the experience era.

Abstract: An online experiment was conducted to test the effect of branded entertainment on the consumers' brand image. Implicit and explicit brand image were measured after exposure to branded entertainment, commercials and no marketing. Implicit brand image was measured through an evaluative priming task and explicit brand image was measured through a questionnaire.

While no effect of marketing strategy was found on explicit brand image, consumers who were exposed to regular commercials had a more positive implicit brand image compared to no exposure to marketing. Branded entertainment, however, seemed to have a more specific effect; only young women and older men seemed to have a more positive implicit brand image, while this effect did not occur for young men and older women. It is concluded that branded entertainment is only effective when exposed to the intended target group, while regular commercials are effective in a larger population.

Location project: Bloakes
External supervisor: Guus van Loon
UU supervisor: Ben Harvey

Name: Mitchel Kappen
Project Title: Is it all in the name? Measuring Implicit Brand Associations for Company Logo's With and Without Company Name

Abstract: Brand logos play a crucial role in a company's marketing communication as it is one of the first elements of a brand consumers are exposed to (Schechter, 1993). It is therefore of no surprise that companies shift to using more billboards displaying merely a name and a logo, rather than more traditional advertisements in magazines (Henderson & Cote, 1998). As people's taste changes, so do trends in design. Companies update their logo with small changes, to keep up with trends and keep their image fresh, whilst maintaining familiarity among their consumer base (Walsh, Page, Winterich, & Mittal, 2010). The current trend in (logo) design is a rather minimalist one in which anything abundant is removed. This happens to such an extent, that some companies have taken their company name out of their logo altogether. This research will use an alteration of the Brand Association Reaction Time Task to test whether people have different associations with a brand when a company logo is included or excluded from the logo. Moreover, we expect to find an influence of one's familiarity with the brand, and one's commitment to the brand.

Location project: Alpha.One
External supervisor: Rosanne van Diepen
UU supervisor: Joris Elshout

Name: Nathaniël Haanschoten
Project Title: Good vibrations: communicating vibrotactile patterns to individuals with deafblindness using a wearable subwoofer on the wrist.

Abstract: For individuals with deafblindness status checking of household devices requires a lot of effort (Hagenaars 2018). Since vibrotactile technology is ubiquitous and individuals with deafblindness have a significant better sense of passive touch, the present experiment tests whether tactons delivered through a smartwatch on the wrist could improve information access, which for people with deafblindness remains

limited to the immediate surroundings. Previous research (Cauchard et al. 2016; Hagedaars 2018) suggests adding a pre-signal to tactons could make status-checking more efficient. However, it is not yet known whether this addition adds to satisfaction while simultaneously maintaining the recognition rate. In a within-subjects experiment ten participants, both deafblind (n=6) and sighted-hearing (n=4), were asked to learn seven different tactons. During the making of a haptic puzzle, the number of correctly recognized tactons and after each condition (that is with and without pre-signal) tacton satisfaction was recorded. It was hypothesized that there would be a significant difference on tacton satisfaction between conditions and that there would be no significant difference on recognition score between conditions. Preliminary results show that there is neither a significant effect of pre-signal on tacton satisfaction nor on the number of correctly recognized tactons. An explanation is that some participants didn't mind while others did find it more satisfying to have a pre-signal and a couple of them were even more satisfied without one. It is therefore advised to include a function on the smartwatch to turn the pre-signal on and off. Concerning the non-significant difference between recognition scores, it is possible that the same applies for the tacton satisfaction, namely that some people benefitted from the pre-signal while for others it was confusing. For future research it is recommended to let people with deafblindness choose which tactons they would like to associate to their household devices and don't use preprogrammed tactons as was the case in the present experiment.

Key words: Vibrotactile perception, deafblindness, assistive wearable

Location project: Bartiméus Doorn

External supervisor: Mijkje Worm

UU supervisor: Krista Overvliet

Name: Niilo Valtakari

Project title: The intricacies of dual eye tracking: Building and testing a setup

Abstract: Technological advancements in the field of eye tracking have made it possible to employ the simultaneous use of two eye trackers to investigate human interaction in greater detail. Various researchers have used this type of a dual eye tracking approach to examine gaze behavior during face-to-face interaction. It is not yet clear what type of eye tracker and what type of setup is most feasible for this purpose. The goal of this internship was to build a dual eye tracking setup that can be used to conduct experiments investigating interaction between infants and their caregivers. We aimed to build a one-computer setup that is both flexible and easy to use. The setup uses two Tobii Pro Nano eye trackers and two Logitech BRIO webcams. Participants are seated opposing each other with one camera providing a view of what each participant is seeing. Gaze coordinates are mapped onto camera recordings. The setup is operated using a Python script that takes advantage of multithreading in conjunction with the computer vision library OpenCV. The end product of the internship is a working setup accompanied by a detailed report describing the setup and its components, along with instructions on how to operate it.

Location project: Utrecht University

External supervisor: N/A

UU supervisor: Ignace Hooge

Name: Rob Koppenaal

Project Title: Reducing noise complaints in an open office

Abstract: Noise problems can be a serious issue for companies with an open office concept. Background noise causes distraction and concentration problems. This results in extra breaks, work of lower quality and exhaustion. This research focused on the question: "Could a poster intervention or a discussion group reduce the noise complaints for Stads Kantoor Utrecht?"

In the first- and third week, noise levels were measured in decibels and the opinions of the co-workers got established with questionnaires (199 participants). Two different interventions took place on two different floors. On one floor the researcher led a discussion group (20 participants) and on the other one, posters were placed on every work spot.

The two interventions did not contribute to reduce the noise complaints.

Location project: Stads Kantoor Utrecht

External supervisor: Wouter Dijkmans

UU supervisor: Joris Elshout

Name: Sjors van de Ven

Project Title: In search of the perfect fit: improving hiring decisions based on objective psychophysiological biometrics and artificial intelligence

Abstract: Background: A good fit between employee and employer is of high importance for both parties to increase the loyalty, quality of the output, the life satisfaction, motivation and general health of the employee. However, hiring decisions are often based on irrelevant subjective perceptions such as gender or race. Therefore, the demand for objective, data-driven hiring advice is increasing. Objectives: A custom algorithm has been developed that opts to give hiring device based on objective psychophysiological biometrics and artificial intelligence. One of the biometrics is the heart rate, which has not been scientifically validated yet. Therefore, the aim of this study is to investigate the resemblance of the output of the heart rate detecting algorithm and a valid heart rate monitor device. Subjects: Participants were recruited through personal contacts and the Social and Behavioral Sciences research participation system of Utrecht University. Methods: Participants executed an online application assessment, during which the heart rate was measured with a Polar H7 heart rate monitor worn on a chest strap and simultaneously with remote photoplethysmography (rPPG). Statistics: The correlation between the heart rate measured with a Polar H7 heart rate monitor and with rPPG will be investigated with the Pearson's correlation coefficient. The variables were the average heart rate, the standard deviation of successive differences, the root mean square of successive differences, the percentage of successive intervals differing from each other by more than 50 ms and the ratio of the low frequency and high frequency content of the interval data. Expected results: Assuming the algorithm works properly, a high positive correlation is expected to be found between the two methods.

Location project: Neurolytics

External supervisor: Felix Hermesen

UU supervisor: Krista Overliet

Name: Sophia Eijnsman

Project Title: A predictive model for the long term liking of bitter

Abstract: Earlier studies have investigated the innate aversion towards the liking for bitter substances/food. Within this study, we investigated the liking for bitters, whether it is possible to increase liking for bitter taste through conditioning and if it is possible to distinct between different bitter flavors.

Location project: Wageningen Research & University

External supervisor: Sanne Boesveldt

UU supervisor: Stefan vd. Stigche

Name: Willem de Bruijne

Project Title: Validation of the FocusFit neurofeedback device: Applicable for altering attentional states?

Abstract: Neurofeedback as a way to increase attentional processes has gained a lot of interest in the recent years. The FocusFit headset (BrainCo, Boston) was designed as an easy-to-use and affordable product to help its users attention getting primed before a workout, but it can also be used as support for studying in classrooms. With the aid of neurofeedback at mental preparation games and guided meditation sessions,

attentional levels are stated to be controlled superiorly. However, the detection of a users attention level via a consumer grade electroencephalogram (EEG) can be quite challenging, since it depends on numerous higher cognitive functions. Therefore, the aim of this study is to assess the quality of the data of this headset during meditation and several attentional demanding tasks. 15 participants will be measured while performing a meditation session and several tasks at different attention levels. The average power of the frequency bands that can be extracted from the FocusFit will be used for the comparison of the different attention levels.

Location project: BrainFitness

External supervisor: Harm Bavinck

UU supervisor: Larissa Meijer

Name: Willem van der Maden

Project Title: What Good Are Twenty-five Positive Emotions? An Extension Of The Emotion Typology

Abstract: This thesis aims to generate knowledge necessary in order to expand the negative emotion typology with positive emotions. Knowledge of discrete emotions is called emotional granularity. Individuals with high emotional granularity have been proven to exhibit other psychological characteristics, therefore the relation between individuals with high positive emotional granularity and creativity has been tested. A list of 27 positive emotions, created by emotion experts, has been substantiated by psychological literature. In order to validate the list of positive emotions, 25 of the 27 positive emotions have been tested in a movie clip validation study (n = 179, 60% male). Accordingly, in order to test the relation between individuals with high positive emotional granularity and creativity. The same participants completed an Alternate Uses Task. The results showed that all tested positive emotions have been validated. For the relation between individuals with high positive emotional granularity and creativity, there was a medium correlation meaning that individuals who correctly identify emotions in video clips, perform better on a divergent thinking task.

Location Project: Utrecht University

External supervisor: N/A

UU supervisor: Joke Baas